

Intrusion Prevention System

Name	Tamim Haider
Project Name	Intrusion Prevention System (IPS)
Attacker OS	Purple Linux
Victim OS	Kali Linux

Step 1: Check the Victim's Network Interface

On the Victim VM:

```
ip -br addr  
ip route
```

Interface in use: wlan0

Step 2: Check the Attacker's IP

On the Attacker VM:

```
ip -br addr  
  
Attacker: 192.168.0.103  
Victim: 192.168.0.107
```

Test connectivity from the attacker:

```
ping 192.168.0.107
```

Step 3: Make Sure Suricata Is Installed

On the Victim:

```
suricata --build-info
```

Version: Suricata 8.0.6

Check the rules directory:

```
sudo ls -l /var/lib/suricata/rules/  
  
classification.config  
suricata.rules  
local.rules
```

Step 4: Create the Test IPS Rule

```
sudo nano /var/lib/suricata/rules/local.rules
```

Rule content:

```
drop icmp any any -> any any (msg:"LAB TEST ICMP BLOCK"; sid:1000001; rev:2;)
```

Save: Ctrl + O → Enter → Ctrl + X

Verify:

```
sudo cat /var/lib/suricata/rules/local.rules
```

Step 5: Make Sure Suricata Loads local.rules

```
sudo nano /etc/suricata/suricata.yaml
```

Confirm rule-files section:

```
rule-files:  
- suricata.rules  
- local.rules
```

Step 6: Test the Suricata Configuration

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

Expected output: "Configuration provided was successfully loaded." with the local rule (1000001) loaded.

Step 7: Clear the Old Suricata Logs

```
sudo truncate -s 0 /var/log/suricata/fast.log  
sudo truncate -s 0 /var/log/suricata/eve.json  
  
sudo cat /var/log/suricata/fast.log
```

Step 8: Add NFQUEUE to the Victim Firewall

```
sudo iptables -I INPUT -j NFQUEUE --queue-num 0  
  
sudo iptables -L INPUT -n -v --line-numbers  
  
1 ... NFQUEUE all -- 0.0.0.0/0 0.0.0.0/0 NFQUEUE num 0
```

Step 9: Start Suricata in IPS Mode

Do not use (this is IDS/live-capture mode):

```
sudo suricata -i wlan0 -c /etc/suricata/suricata.yaml
```

Use instead:

```
sudo suricata -c /etc/suricata/suricata.yaml -q 0
```

Expected: "Suricata version 8.0.6 RELEASE" ... "Engine started." Leave this terminal running.

Step 10: Test from the Attacker

```
ping 192.168.0.107  
  
PING 192.168.0.107 ...  
From 192.168.0.103 ...  
Request timeout  
Request timeout  
Request timeout
```

Step 11: Check Suricata's IPS Output

Stop Suricata on the victim: Ctrl + C

```
nfq: (RX-NFQ#0) Treated: Pkts ...  
nfq: (RX-NFQ#0) Verdict: Accepted ...  
nfq: (RX-NFQ#0) Verdict: ... Dropped ...  
  
Treated: Pkts 167  
Verdict: Accepted 27, Dropped 140, Replaced 0
```

Step 12: Check the Suricata Alert

```
sudo cat /var/log/suricata/fast.log  
  
[**] [1:1000001:2] LAB TEST ICMP BLOCK [**]
```

Search specifically:

```
sudo grep "LAB TEST ICMP BLOCK" /var/log/suricata/fast.log
```

Step 13: Check eve.json

```
sudo grep "LAB TEST ICMP BLOCK" /var/log/suricata/eve.json  
  
sudo tail -n 20 /var/log/suricata/eve.json
```

Step 14: Check the Firewall Packet Counter

```
sudo iptables -L INPUT -n -v --line-numbers  
  
1  50  4200  NFQUEUE  all  --  0.0.0.0/0  0.0.0.0/0
```

Flow confirmed: Packet → INPUT → NFQUEUE → Suricata

Step 15: Complete IPS Test Flow

```
ATTACKER VM (192.168.0.103)  
  | ICMP  
  ▼  
VICTIM VM (192.168.0.107)  
  |  
  INPUT  
  |  
  NFQUEUE 0  
  |  
  SURICATA IPS  
  |  
  Rule matches: LAB TEST ICMP BLOCK  
  |  
  DROP
```

Step 16: Stop the IPS

Stop Suricata: Ctrl + C

```
sudo iptables -D INPUT -j NFQUEUE --queue-num 0  
sudo iptables -L INPUT -n -v --line-numbers
```

Step 17: Start IPS Again Later

Terminal 1, Victim:

```
sudo iptables -I INPUT -j NFQUEUE --queue-num 0  
sudo suricata -c /etc/suricata/suricata.yaml -q 0
```

Terminal 2, Attacker:

```
ping 192.168.0.107
```

Terminal 1, after stopping Suricata:

```
sudo cat /var/log/suricata/fast.log  
sudo iptables -L INPUT -n -v --line-numbers
```

Step 18: Full Command Sequence (Summary)

Victim:

```
ip -br addr  
ip route  
sudo cat /var/lib/suricata/rules/local.rules  
sudo suricata -T -c /etc/suricata/suricata.yaml  
sudo truncate -s 0 /var/log/suricata/fast.log  
sudo truncate -s 0 /var/log/suricata/eve.json  
sudo iptables -I INPUT -j NFQUEUE --queue-num 0  
sudo iptables -L INPUT -n -v --line-numbers  
sudo suricata -c /etc/suricata/suricata.yaml -q 0
```

Attacker:

```
ping 192.168.0.107
```

Back on Victim:

```
sudo cat /var/log/suricata/fast.log  
sudo grep "LAB TEST ICMP BLOCK" /var/log/suricata/eve.json  
sudo iptables -L INPUT -n -v --line-numbers  
sudo iptables -D INPUT -j NFQUEUE --queue-num 0
```